

# The Fertility Dividend: Family Size and the Decline in Child Poverty

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## Abstract

Child poverty under the Supplemental Poverty Measure fell 20 percentage points between 1967 and 2024, a decline conventionally credited to income growth and changing taxes and transfers. I show that 5.4 points—27% of the decline or 3.7 million fewer children in poverty in 2024—instead reflect a fertility dividend: smaller families spread finite resources across fewer children. Because equivalence scales and decompositions condition on family size, standard accounting records demographic change as income growth. I quantify the dividend with a rank-preserving transport of the family-size distribution, and corroborate the mechanism by showing that twin-induced parity transitions still raise child poverty 3–7 points.

**Keywords:** Child poverty; family size; fertility; equivalence scales; poverty measurement

**JEL Classification:** I32, J13, J11

## Introduction

Child poverty in the United States fell by about 20 percentage points between 1967 and 2024 under the anchored Supplemental Poverty Measure. The standard account credits growth in market income and changing taxes and transfers (Burkhauser et al. 2024; Fox et al. 2015; National Academies of Sciences, Engineering, and Medicine 2019; Wimer et al. 2016). I show that 27% of the decline—5.4 percentage points, or roughly 3.7 million fewer children in poverty in 2024—instead reflects declining fertility: as families had fewer children, resources were spread across fewer children, reducing child poverty even holding household resources fixed. The remainder reflects income growth and taxes and transfers, partially offset by rising single parenthood.

This “fertility dividend” is easy to miss in conventional poverty accounting. Standard poverty trend decompositions condition on family size (e.g., Burkhauser et al. 2024; Fox et al. 2015; Meyer and Sullivan 2012; Wimer et al. 2016), and poverty statistics convert household resources into resources per “equivalized” person using equivalence scales. When family size falls and resources are held constant, equivalized income rises by construction, and demographic change is recorded as income growth rather than as a separate driver of child poverty. The resulting dividend echoes the macroeconomic demographic dividend (Bloom et al. 2003; Bloom and Williamson 1998), but the mechanism here operates within families: fewer children share a given amount of family resources, raising resources relative to the poverty threshold. Without separating this channel, accounts of anti-poverty progress overstate the role of income growth and understate the role of demographic change.

A key mechanism through which family size affects poverty is direct resource dilution, the quantity–quality channel emphasized by Becker (1960). An additional child mechanically raises a family’s poverty threshold—its needs standard—and spreads family resources across more children. Unless family resources rise enough to offset the higher threshold,

the family moves closer to poverty. Evidence from higher-order fertility shocks provides little reason to expect such an offset: additional children tend to reduce, not increase, mothers' labor supply (Aaronson et al. 2021; Angrist and Evans 1998).

To quantify the family-size channel, I construct counterfactuals that replace the distribution of the number of children in one year with that of another, holding all else fixed. Within marital-status cells, each mother is reassigned the number of children at the same percentile of the target-year distribution. This rank-preserving transport (Novosad et al. 2022) changes family size and, with it, the poverty threshold and child exposure weights, while leaving household resources and other characteristics unchanged. I then recompute poverty under the counterfactual distribution. The approach relies on a rank-preservation assumption: within marital-status cells, mothers retain their ordinal position in the family-size distribution as that distribution shifts across years. I assess sensitivity to this assignment rule using alternative transport cells and rank orderings, and use twin-birth instruments as an independent check on the mechanism.

The resulting 5.4-percentage-point estimate should be interpreted as the direct resource-dilution effect of fertility decline, not the total effect of fertility change on child poverty. It excludes broader behavioral and compositional channels through which lower fertility may have reduced poverty, including delayed childbearing, higher parental earnings, and greater human-capital investment (e.g., Bailey 2006, 2013; Goldin and Katz 2002). Those channels are not ignored in the decomposition: to the extent that they raise observed family resources, they enter through measured income growth rather than through the family-size channel.

For context, I compare the fertility contribution with the effects of changes in marriage, market income, and taxes and transfers using a Shapley decomposition. Sensitivity analysis varying the transport specification and ranking rule yields estimates ranging from  $-2.3$  to  $-8.5$  percentage points.

As independent evidence on the resource-dilution mechanism, I use twin births as an in-

strument for parity transitions (Angrist and Evans 1998; Rosenzweig and Wolpin 1980). Twin-induced increases in family size raise child poverty by 3–7 percentage points in recent data and 8–16 percentage points in 1960. The IV estimates also reveal the underlying resources-to-needs comparison: additional children raise the poverty threshold substantially, while generating little change in total family income. This pattern supports the interpretation that the poverty effect operates through resource dilution. Extending the analysis to 1900–2023 suggests that fertility was likely a sizable contributor to poverty changes earlier in the century. The mechanism remains active. What changed after 1980 was not the mechanism but the margin—family size among parents stopped falling, so the dividend has largely been collected, and today’s fertility decline operates increasingly through childlessness and fewer first births (Kearney and Levine 2025; Kearney et al. 2022), a shift that changes the number of children rather than resources per child. Policies that succeed in raising births among existing parents could, absent offsetting resources, mechanically raise child poverty. This finding provides a family-level analog to Weil (2026)’s observation that raising U.S. fertility back to replacement would lower living standards for several decades.

The paper makes three contributions: it quantifies the family-size contribution to child poverty over the past several decades; it shows that standard equivalence-scale accounting absorbs this demographic change into “income growth”; and it validates the result with twin-birth instruments.

## **Data**

This paper draws on two complementary microdata sources. The main analysis uses the Anchored Supplemental Poverty Measure (SPM) linked to the Current Population Survey (CPS), 1967–2024 (Flood et al. 2022; Waldfogel et al. 2017). SPM resources equal cash income plus in-kind transfers and refundable tax credits, minus taxes and necessary expenses; thresholds vary with family size and geography.

For the instrumental variables analysis and the long-run extension to 1900, I use Decennial Census and American Community Survey microdata from IPUMS-USA (Ruggles et al. 2023), which provide the sample sizes needed to identify twin births and consistent measures of family structure over time. These datasets report poverty using the Official Poverty Measure (OPM), which compares pre-tax cash income to family-size thresholds. The resource-dilution channel is relevant in each of the poverty metrics used.

The unit of analysis is the child. The sample includes all children ages 0–17 in family units containing at least one adult and one child. Each child is linked to a parent by first matching to a mother (using the IPUMS mother pointer), then to a father if no mother link is available, and finally to the best-available adult in the same SPM resource unit as a guardian. This hierarchy ensures that every child in the sample is linked to an adult whose characteristics (age, marital status, family income) may be used in the analysis. The CPS–SPM sample contains about 2.8 million child-year observations spanning 1967–2024; the Census/ACS sample used for the long-run analysis contains about 14 million person-year records spanning 1900–2023.

For the decomposition, I collapse to one observation per linked adult (henceforth “mother” for exposition) and re-aggregate to children using child weights equal to the mother’s survey weight multiplied by the number of children; this avoids double-counting resources within families while keeping the child as the unit of interest. Poverty is measured using SPM poverty and market poverty (pre-tax, pre-transfer income below the same threshold) to separate market-income growth from transfer expansion.

Family size is the number of co-resident linked children ages 0–17, capped at seven. For the twin-birth IV analysis, I use larger samples—a 5% extract from the 1960 Census and pooled ACS samples from 2015–2024—restricted to mothers aged 30 and older. Twin births are identified as instances in which two children of the same mother share a birth year and quarter; triplets and higher-order multiples are excluded. Appendix Table E1 reports summary statistics for 1967 and 2024.

## Trends in Fertility, Income, and Poverty

Declining fertility has created a growing wedge between the economic experience of households and that of children. Figure 1 compares cumulative growth in median family income and median income per child since 1967. Median family income rose 73%, but median income per child rose 140%—nearly twice as fast. By contrast, income per equivalent person—using the scale embedded in poverty assessment—grew much more slowly because equivalence scales assume steep economies of scale and mute the gains from smaller families. This wedge is the fertility dividend in levels: dividing resources among fewer children raised per-child income directly, while the equivalence-scaled series—the one poverty measurement uses—records only a muted version of that gain. Conventional metrics therefore understate the improvement in children’s material living standards.

The mechanism linking fertility decline to poverty reduction operates through a steep and persistent poverty–family size gradient. In 2023, about 10% of children in one-child families lived in poverty compared to about 25% of children in four-child families—a ratio of nearly 2.5 to 1. This gradient has been remarkably stable across decades: using Census data from 1950 to 2023, the relative poverty disadvantage of large families persists despite enormous changes in overall poverty levels (Appendix Figure D1, Panel (b)). Being in a large family is a relative risk factor for child poverty comparable in magnitude to having an unmarried parent or a less-educated mother. Over the twentieth century, the share of children living in families with four or more children collapsed from over half to just 15%, shifting a large fraction of the child population from high-poverty large families to lower-poverty small families.

Because I hold household resources fixed, I exclude behavioral channels and isolate the direct effect of family size. Those excluded channels reinforce rather than offset the result: the fertility transition was accompanied by later childbearing and increases in maternal education, labor force participation, and earnings, which also reduced child poverty (Bailey

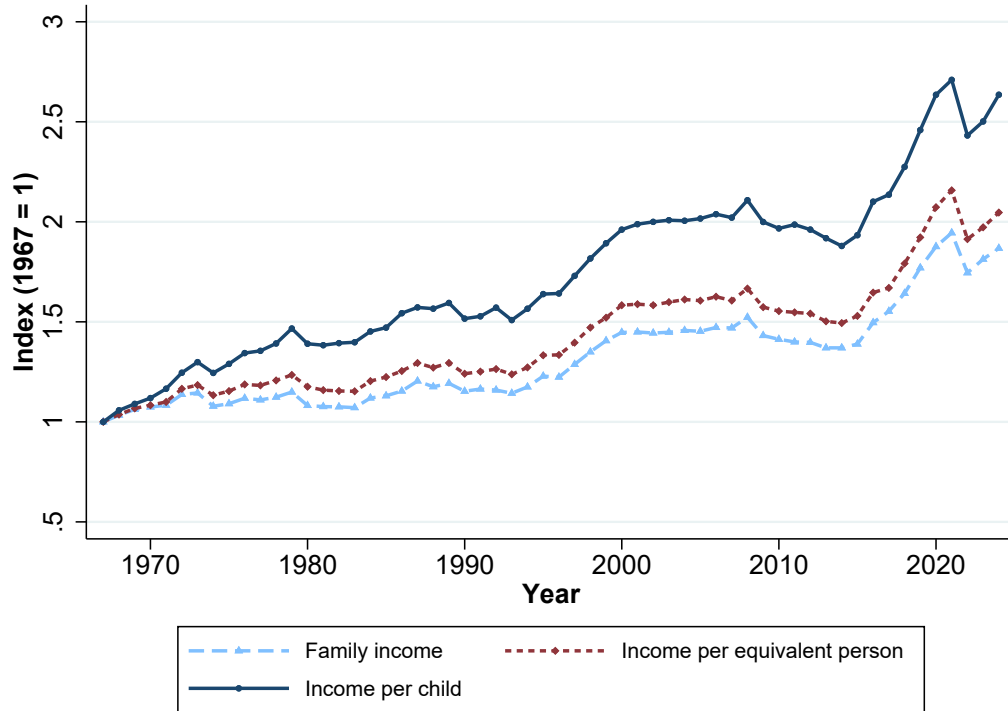


Figure 1: Income per child outpaced family income as families shrank, 1967–2024

*Notes:* Figure plots cumulative growth in three income measures among families with children, indexed to 1967. Family income (dashed) measures total SPM resources. Income per equivalent person (short-dashed) applies the Betson three-parameter equivalence scale used in the SPM (Betson 1996). Income per child (solid) divides family income by the number of children. The divergence between family income and income per child reflects the fertility dividend; the gap between income per child and the equivalized series measures the per-child gains that equivalence scaling conceals.

2013). Early childbearing is heavily selected on low economic opportunity, and delaying first births appears to do little to raise mothers’ long-run resources (Kearney and Levine 2012), so parental resources look relatively inelastic to fertility timing even as the threshold and resources per child respond mechanically to the number of children.

The cross-sectional association between family size and child poverty could reflect selection—systematic differences in income and other characteristics across families with different fertility—rather than a causal effect of additional children. The decomposition and the twin-birth IV analysis address this in complementary ways: the decomposition constructs counterfactual poverty rates that hold the income distribution fixed while varying family size, isolating the mechanical “resource dilution” component; the IV analysis uses twin births to generate quasi-random parity transitions, testing whether an additional child raises poverty even absent selection.

## **Decomposition of Poverty Change**

I construct counterfactual scenarios that ask: what would child poverty have been if families had had fewer (or more) children, but household resources stayed the same? I replace each year’s family-size distribution with that of another year and recompute poverty, holding resources fixed. I decompose the reduction in child poverty between 1967 and 2024—approximately 20 percentage points—into contributions from fertility, marriage, market income, and taxes and transfers.

The central counterfactual-assignment problem is how to attach the target-year family-size distribution to the reference-year population of families and resources. The aggregate decline in family size is observed, but it does not by itself say whether the counterfactual change should be assigned to higher- or lower-resource families. Because the poverty effect can depend on that assignment, I impose a rank-preservation assumption: within marital-status cells, mothers who are relatively high-fertility in one year remain relatively high-fertility as the family-size distribution shifts. This allows fertility to be correlated with



income and other characteristics; it requires only stability of ordinal fertility ranks within cells.

For example, if the 80th-percentile married mother has four children in 1967 but three in 2024, I assign the 1967 mother three children, adjust her poverty threshold for the smaller family (via the implied change in equivalence scales), and count three (not four) children as exposed to her household’s poverty status when aggregating to child poverty (i.e., mother weights are multiplied by counterfactual children). Because family size is discrete, within-tie ranks are not identified. I break ties randomly in the main specification and report bounds using income-ascending and income-descending within-tie orderings, following the partial-identification logic in Manski (2003). The counterfactual changes only the family-size distribution (and therefore poverty thresholds and child exposure weights). It does not change family resources, marital status, geographic threshold adjustments, or any other characteristic.

Formally, within marital-status cell  $m$ , I compute each mother’s weighted percentile  $p_i$  in the reference-year distribution of the number of children and assign the counterfactual number of children  $C_i^{\text{cf}} = \min\{c : F_m^{t_1}(c) \geq p_i\}$ , where  $F_m^{t_1}$  is the target-year weighted empirical CDF. I then recompute child poverty using the counterfactual threshold

$$T_i^{\text{cf}} = T_i \times \frac{\text{scale}(A_i, C_i^{\text{cf}})}{\text{scale}(A_i, C_i)}, \quad (1)$$

where  $\text{scale}(A, C) = (A + 0.5C)^{0.7}$  for  $A \geq 2$  and  $(1.3 + 0.5C)^{0.7}$  for  $A = 1$  (single-adult units with children). This ratio-based rescaling changes only the family-size component of the threshold; geographic and other adjustments are held fixed. Child poverty is re-aggregated using counterfactual child weights equal to the mother’s survey weight times  $C_i^{\text{cf}}$ .

This procedure is a rank-preserving distributional transport (Novosad et al. 2022). Let  $\theta_i$  denote an unobserved, time-invariant fertility propensity for mother  $i$  and let  $X_i$  index the marital-status cell. The identifying assumption is that the conditional distribution of  $\theta_i$  is

stable over time:

$$\Pr(\theta_i \leq \theta \mid X_i = x, t_0) = \Pr(\theta_i \leq \theta \mid X_i = x, t_1) \quad \text{for all } \theta, x. \quad (2)$$

Under this assumption, a mother at percentile  $p$  of the family-size distribution within cell  $x$  in  $t_0$  is also at percentile  $p$  in  $t_1$ , so transporting by matching percentiles recovers the counterfactual assignment. In plain terms: mothers who tend to have more children than their peers in one era would also tend to have more children in another. This is weaker than assuming fertility is independent of income: high-fertility mothers may be poorer, so long as their relative fertility ranks are preserved as the distribution shifts. Table 1 shows robustness to alternative rank-income mappings; expanding the transport cells to race  $\times$  marital status yields a similar estimate (Appendix B); and the twin-birth instruments introduced below provide corroborating evidence that does not rely on rank stability.

To benchmark the fertility contribution, I compare it with changes in marriage, market income, and taxes and transfers using a Shapley decomposition (Shapley 1953; Shorrocks 2013; Young 1985), which attributes the overall change in poverty across these factors by averaging each factor’s marginal contribution across all possible orderings.

To implement this comparison, I begin with a mother-level dataset that keeps one observation per mother and aggregate outcomes to children using child weights equal to the mother’s survey weight times the current number of children. For each ordering of factors, I start from the reference-year population and update one component at a time, re-computing child poverty after each step. Fertility is updated using the rank-preserving transport described above: within marital-status cells, mothers are ranked by their current number of children and reassigned the number of children at the same percentile of the target-year distribution. Holding resources fixed, this changes child poverty through two channels only: the poverty threshold is adjusted to reflect the counterfactual number of children, and the child-level aggregation weights change accordingly. The marriage com-

ponent is a two-point inverse propensity weight update that matches the target-year child-weighted marriage share without changing mothers’ observed characteristics; after fertility and marriage are applied, the remaining change is allocated between market income and taxes and transfers by comparing market poverty with total poverty under the same size-adjusted threshold. Averaging each factor’s marginal contribution across orderings yields components that sum exactly to the total change in child poverty.

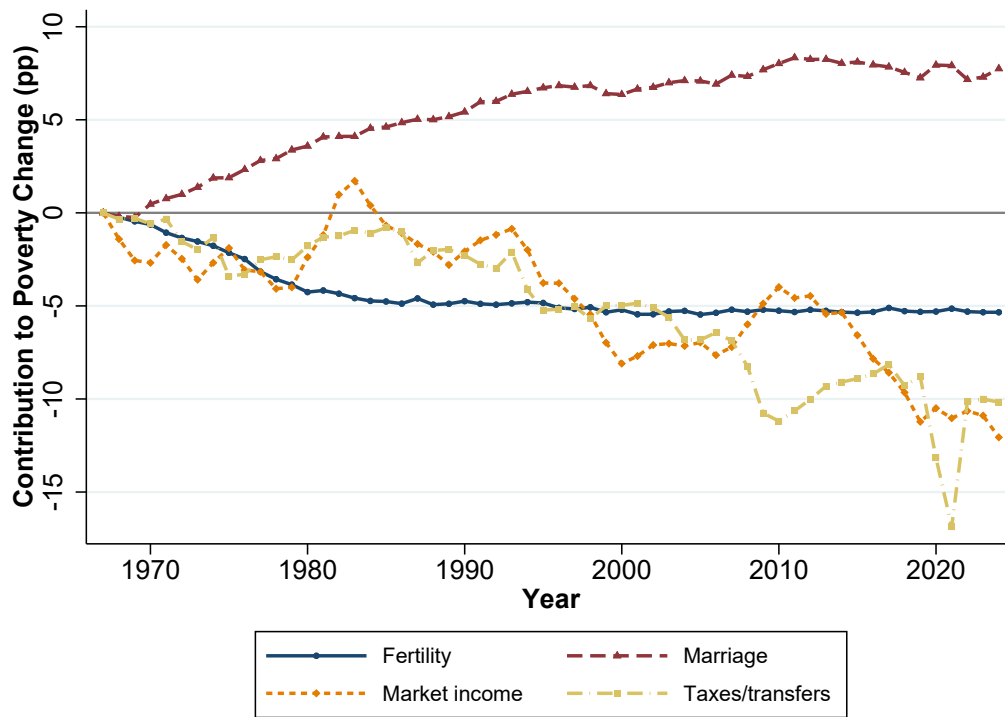


Figure 2: Fertility decline accounts for 27% of the 1967–2024 fall in child poverty

*Notes:* Figure plots cumulative Shapley contributions of each factor to the change in child poverty (SPM) relative to 1967. The fertility component captures changes in family size within marital-status cells; components sum to the total change by construction.

Figure 2 presents the cumulative results. Fertility decline reduced child poverty by 4.3 percentage points between 1967 and 1980 alone—when family size among parents fell fastest—reaching  $-5.4$  points by 2024, 27% of the total decline; the contribution was essentially complete by 2000 and flat thereafter.<sup>1</sup> About 90% of this fertility contribution reflects re-

<sup>1</sup>Decomposition totals reflect mother-level aggregation on the estimation sample, which differs slightly from raw child-level rates mostly because of slight differences in within-family sample weights:

source dilution: having fewer children lowers the poverty threshold facing a given family and therefore makes it less likely to be poor. The remainder reflects child-weighting: as family size falls, children are reallocated away from larger, higher-poverty families and toward smaller, lower-poverty families. Fertility decline among married mothers accounts for roughly half of the effect, with unmarried mothers contributing disproportionately relative to their population share in 1967 due to steeper poverty–fertility gradients. Marriage decline increased poverty by 7.7 pp, while market income growth contributed  $-12.0$  pp and transfer expansion  $-10.2$  pp.

Table 1 reports the fertility component from the full Shapley decomposition under alternative transport designs and tiebreaking rules. In every specification, the overall decomposition is the same: poverty is decomposed into fertility, marriage, market income, and taxes and transfers; marriage is handled by reweighting to match the target-year child-weighted marriage share; and market income versus taxes and transfers are separated in the second stage using market poverty and total SPM poverty under the same size-adjusted threshold. What varies across rows and columns is only the way the fertility counterfactual is constructed. The rows vary the conditioning set and treatment of fertility timing. The main specification (A) transports the number of children within marital-status cells. Method B instead applies an unconditional transport that ignores marital status and matches only the marginal distribution of children. Method C adds fertility timing by first aligning age at first birth and then transporting the number of children within finer cells defined by marital status and age-at-first-birth quintile. A+IPW provides a timing-adjusted version of the main specification by first reweighting the reference-year sample to match the target-year age-at-first-birth distribution and then transporting the number of children within marital-status cells. The columns vary how mothers are ordered within family-size mass points, where the rank-preserving transport is otherwise not unique because

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the decomposition sums to  $-19.8$  pp (baseline 29.4%) while the simple child-level difference is  $-20.1$  pp (29.7%  $\rightarrow$  9.6%). The Shapley baseline of  $-5.4$  pp averages the fertility contribution over orderings; the single-transport A estimate in Table 1 ( $-5.3$  pp) isolates fertility alone and differs from the Shapley number only through the marriage-ordering interaction.

the number of children is discrete. Because many mothers share the same observed number of children, their within-cell fertility ranks are only partially observed. The income-descending and income-ascending tiebreakers assign these latent ranks in the two extreme monotone orderings with respect to income; they thereby provide lower and upper bounds on the estimated fertility contribution. The random and copula tiebreakers instead yield interior estimates under weaker assumptions about the relationship between income and latent fertility rank. Appendix B describes the construction of these specifications in detail.

Table 1: Sensitivity of the estimated fertility contribution to child poverty

| Transport method                     | Tiebreaker |        |          |        |
|--------------------------------------|------------|--------|----------|--------|
|                                      | Inc-desc.  | Random | Inc-asc. | Copula |
| A: Within marriage                   | −2.9       | −5.3   | −8.5     | −5.6   |
| B: Marginal                          | −2.3       | −5.0   | −8.1     | −5.2   |
| C: Sequential AFB $\rightarrow$ fert | −2.5       | −4.9   | −7.6     | −5.9   |
| A+IPW: Marriage + AFB                | −3.8       | −6.1   | −8.2     | −6.6   |

*Notes:* Table reports the fertility-transport contribution to child poverty (percentage points, 1967–2024) under four conditioning specifications and four within-tie ordering rules. Income-descending assigns counterfactual children to higher-income mothers first (lower bound); income-ascending assigns to lower-income mothers first (upper bound). The single-transport estimate for the main specification (A, random tiebreaker) is −5.3 pp; the −5.4 pp figure cited in the text is the Shapley fertility contribution, which adds the marriage-ordering interaction. Source: CPS ASEC, 1967–2024.

Across all 16 method×tiebreaker combinations, the estimated contribution ranges from −2.3 to −8.5 percentage points, with the central estimates (random and copula tiebreakers) clustering between −4.9 and −6.6 pp. The copula tiebreaker uses a Gaussian copula to impose the observed within-cell rank correlation between income and family size when ordering mothers within mass points; it produces estimates close to the random tiebreaker because the income–fertility gradient within cells is relatively flat.

The final row of Table 1 (A+IPW) is a timing-adjusted variant of the main transport. Before transporting family size within marital-status cells, I reweight the reference-year sample using inverse probability weights so that the age-at-first-birth distribution matches the target year, and then recompute counterfactual poverty. Relative to A, this adjustment

changes the estimated fertility contribution by about 1 percentage point. Because the calculation uses the reference year’s incomes, age–earnings profile, and returns to education and work, it is not a causal estimate of how delayed childbearing affected earnings or human capital. Rather, it shows that shifting the age-at-first-birth distribution toward its later target-year pattern is associated with additional poverty reduction even before assigning any behavioral earnings gains to delayed fertility.

The estimated fertility contribution is also robust to alternative poverty measures. Appendix Table B2 recomputes the transport effect under square-root, per-capita, deep-poverty, and fixed per-child thresholds, and under the two equivalence scales economists have estimated directly from expenditure data—Lazear and Michael (1980) and Nelson (1988), which embed the largest economies of scale in the literature (Fernández-Villaverde and Krueger 2007). The Betson and square-root scales yield nearly identical estimates (−5.0 and −4.9 pp); the consumption-based scales bracket this benchmark (−6.0 pp for Lazear–Michael, −3.2 pp for Nelson); and the per-capita measure produces a larger effect (−8.2 pp). The central estimate is thus not an artifact of the SPM’s assumed scale.

The estimated fertility contribution illustrated in Table 1 should be interpreted as a mechanical resource-dilution effect: it captures how smaller families raise resources per child. This effect is economically meaningful: roughly half the size of the entire contribution of taxes and transfers (−10.2 pp). The decomposition does not identify broader channels through which fertility control changed behavior and composition, such as delayed childbearing, higher educational attainment, increased labor force participation, and higher earnings (e.g., Bailey 2006, 2010, 2013; Goldin and Katz 2002). Any income gains arising from these mechanisms are absorbed by the market-income component rather than attributed to fertility per se. This applies directly to the secular rise in parental education, the largest compositional change of the period: because more-educated parents earn more, education’s contribution enters through the market-income component (−12.0 pp), not as an omitted competitor to the fertility channel.

The decomposition holds family resources fixed and asks whether smaller families are less likely to be poor because fewer children share those resources. To corroborate this resource-dilution interpretation, I use twin births as instruments for parity transitions (Angrist and Evans 1998; Rosenzweig and Wolpin 1980). This evidence addresses the concern that the poverty–family-size gradient partly reflects selection into larger families: conditional on a pregnancy at a given parity, a twin birth generates a plausibly exogenous increase in family size.

Parity transitions are defined on observed co-resident linked children ages 0–17, so the estimates use near-completed fertility measured from linked co-resident children rather than literal completed births. Using the 1960 Census and pooled 2015–2024 ACS data, I estimate the effect of crossing each parity threshold—going from one child to two, from two to three, and so on—by instrumenting an indicator for having at least  $n + 1$  co-resident children with a twin birth at pregnancy  $n$  (Angrist et al. 2010; Black et al. 2005), restricting the sample to mothers aged 30 and older with at least  $n$  pregnancies; the age-30 restriction is intended to approximate completed fertility; Appendix Table C4 shows the estimates are nearly identical when the sample is widened to all mothers aged 18 and older. The first stage is strong at all margins, with F-statistics exceeding 3,000.

Table 2 presents the results. Panel A shows that twin-induced parity transitions increase the probability that a child lives in poverty by 3–7 percentage points in the modern period and 8–16 percentage points in the 1960 Census, when baseline poverty rates were higher. Panel B shows that total family income is largely unaffected by twin-induced family-size increases; Panel C shows that earnings per child fall sharply. Panel D shows the other side of the resources-to-needs comparison: an additional child raises the family’s official poverty threshold by 15–36 log points across margins, precisely estimated. The contrast between Panels B and D is the mechanism—the needs standard rises sharply while resources do not—so measured poverty rises. This four-part pattern—poverty rises, the threshold rises sharply, income does not rise, resources per child fall—is the signature of resource dilu-

tion. The additional child divides fixed resources among more children, with no offsetting increase in parental labor supply (Appendix C.7). The pattern also implies the fixed-resource decomposition is conservative: the needs standard rises by more than resources ( $\Delta \log T > \Delta \log R$ ; Panels B and D), so the estimate understates the full poverty effect of larger families.

At higher parities—where much of the historical fertility decline occurred—the OLS and IV estimates are similar, supporting the decomposition’s reliance on the observed poverty–family size gradient at the margins most relevant for aggregate trends, though estimates at the highest observed parities are imprecise due to small samples of mothers at those margins. The sole anomaly is at the  $1 \rightarrow 2$  margin in modern data, where family income rises, perhaps reflecting selection among mothers using assisted reproductive technology; this anomaly is absent in the 1960 Census and at all higher parities. (Full regression results including OLS comparisons are reported in Appendix C.)

The IV estimates are also broadly consistent with the magnitude implied by the decomposition. Mean family size fell from 3.3 to 2.4 children between 1967 and 2024 (a decline of 0.9), implying a substantial shift in births away from higher parities. As a back-of-the-envelope comparison, scaling the modern parity-specific IV estimates (roughly 3–8 percentage points across margins) by the 0.9-child decline implies a poverty reduction of roughly 3–7 percentage points, which brackets the decomposition’s estimate of 5.4 percentage points. The larger 1960 IV estimates reflect a greater mass of families clustered near the poverty threshold in 1960: when more families sit just above the line, an additional child tips a larger share below it—a consequence of the more compressed income distribution near the threshold, not of the higher baseline poverty rate per se. Together, the IV results indicate that the poverty–family size gradient reflects resource dilution rather than selection and support the decomposition’s 5.4-percentage-point estimate.



Table 2: Twin-IV mechanism test: effect of crossing a parity threshold

|                                             | Parity Transition |           |          |          |          |          |
|---------------------------------------------|-------------------|-----------|----------|----------|----------|----------|
|                                             | 1 → 2             | 2 → 3     | 3 → 4    | 4 → 5    | 5 → 6    | 6 → 7    |
| <i>Panel A: Child Poverty (OPM)</i>         |                   |           |          |          |          |          |
| ACS                                         | 0.031             | 0.037     | 0.043    | 0.038    | 0.079    | 0.065    |
|                                             | (0.004)           | (0.004)   | (0.009)  | (0.016)  | (0.037)  | (0.063)  |
|                                             | [< .001]          | [< .001]  | [< .001] | [.017]   | [.033]   | [.306]   |
| Census                                      | 0.099             | 0.084     | 0.095    | 0.126    | 0.143    | 0.158    |
|                                             | (0.011)           | (0.010)   | (0.013)  | (0.018)  | (0.026)  | (0.033)  |
|                                             | [< .001]          | [< .001]  | [< .001] | [< .001] | [< .001] | [< .001] |
| <i>Panel B: Log Family Income</i>           |                   |           |          |          |          |          |
| ACS                                         | 0.115             | 0.014     | −0.011   | 0.039    | −0.154   | 0.043    |
|                                             | (0.010)           | (0.010)   | (0.020)  | (0.032)  | (0.092)  | (0.128)  |
|                                             | [< .001]          | [.173]    | [.595]   | [.225]   | [.093]   | [.738]   |
| Census                                      | 0.017             | −0.031    | −0.026   | −0.008   | −0.060   | −0.103   |
|                                             | (0.020)           | (0.018)   | (0.022)  | (0.031)  | (0.050)  | (0.066)  |
|                                             | [.408]            | [.082]    | [.234]   | [.784]   | [.233]   | [.119]   |
| <i>Panel C: Log Earnings per Child</i>      |                   |           |          |          |          |          |
| ACS                                         | −0.731            | −0.459    | −0.340   | −0.272   | −0.332   | −0.234   |
|                                             | (0.011)           | (0.011)   | (0.021)  | (0.040)  | (0.092)  | (0.140)  |
|                                             | [< .001]          | [< .001]  | [< .001] | [< .001] | [< .001] | [.096]   |
| Census                                      | −1.111            | −0.629    | −0.490   | −0.409   | −0.271   | −0.328   |
|                                             | (0.030)           | (0.023)   | (0.030)  | (0.041)  | (0.058)  | (0.087)  |
|                                             | [< .001]          | [< .001]  | [< .001] | [< .001] | [< .001] | [< .001] |
| <i>Panel D: Log Poverty Threshold (OPM)</i> |                   |           |          |          |          |          |
| ACS                                         | 0.232             | 0.188     | 0.147    | 0.149    | 0.198    | 0.241    |
|                                             | (0.002)           | (0.002)   | (0.003)  | (0.006)  | (0.015)  | (0.020)  |
|                                             | [< .001]          | [< .001]  | [< .001] | [< .001] | [< .001] | [< .001] |
| Census                                      | 0.363             | 0.245     | 0.204    | 0.208    | 0.218    | 0.232    |
|                                             | (0.006)           | (0.005)   | (0.005)  | (0.007)  | (0.010)  | (0.009)  |
|                                             | [< .001]          | [< .001]  | [< .001] | [< .001] | [< .001] | [< .001] |
| Obs. (ACS)                                  | 2,644,308         | 1,511,086 | 507,798  | 146,809  | 42,508   | 14,887   |
| Obs. (Census)                               | 921,213           | 622,001   | 338,051  | 165,357  | 77,334   | 37,721   |

*Notes:* 2SLS estimates. Each column instruments an indicator for having at least  $n+1$  children with a twin birth at pregnancy  $n$ . Panel A shows that twin-induced parity transitions raise child poverty; Panel B shows that total family income is largely unaffected; Panel C shows that earnings per child fall proportionally; Panel D reports the (mechanical) effect on the log official poverty threshold, isolating the change in the needs standard induced by an additional child. The positive income effect at the 1 → 2 margin in the ACS likely reflects ART-related selection; this anomaly does not appear in the 1960 Census (which predates ART) or at higher parities. Controls include maternal age, education, marital status, survey year, and age-at-first-birth fixed effects. Sample restricted to mothers aged 30+. ACS = pooled American Community Survey, 2015–2024; Census = 1960 Decennial Census. OLS comparisons are in Appendix C. Heteroskedasticity-robust standard errors in parentheses; continuous p-values in brackets.

## Long-Run Perspective: 1900–2023

The CPS–SPM series begins in 1967—after much of the twentieth-century fertility transition—and therefore misses decades in which U.S. family size fell more sharply than in the post-1967 period. To place the main results in a longer historical context and to assess robustness to an alternative poverty concept, I extend the analysis to Decennial Census and ACS microdata spanning 1900–2023 (IPUMS USA), using the Official Poverty Measure (OPM). This extension has three aims. First, it quantifies the mechanical implications of fertility change during 1900–1967, when shifts in family size were larger than in the CPS era. Second, it tests whether the same resource-dilution channel appears under the OPM—a pre-tax cash-income measure with different treatment of taxes and in-kind transfers than the SPM. Third, it provides an external check on the CPS-based decomposition in the years the two data sources overlap.

Figure 3 juxtaposes decade-by-decade decompositions from the CPS–SPM analysis (panel a) and the Census/ACS–OPM analysis (panel b). Panel a reports Shapley contributions from fertility, marriage, market income, and taxes/transfers under the SPM; panel b reports fertility and marriage components under the OPM, with a residual that largely reflects growth in cash income (because taxes and non-cash transfers are not used in the OPM).

In panel a, fertility’s poverty-reducing effect is concentrated early in the CPS window. During 1967–1980—when average family size fell rapidly—fertility alone reduced child poverty by 4.3 percentage points, exceeding even the contribution of market income growth. After 1980, fertility’s marginal contribution is small, consistent with the deceleration in family-size decline. The dominant drivers shift accordingly: market income growth drives much of the poverty decline in the 1990s (−5.8 pp) and 2010s (−8.0 pp), while transfer expansion is the primary factor in the 2000s (−6.1 pp). (Appendix Table A1 reports the full numerical results; Figure A1 provides a complementary decomposition of income per

child.)

From 1950 onward, panel b applies the same fertility transport within marital-status cells and the same marriage IPW reweighting under the OPM; the residual absorbs income growth and policy changes. Because OPM poverty is not consistently available before 1950, the 1900–1950 bars are backward-transport implied effects anchored to the 1950 baseline rather than observed poverty decompositions.

Two patterns stand out. First, in the years that overlap the CPS era, the OPM-based estimates mirror the SPM-based timing: fertility’s contribution is concentrated in the late 1960s and 1970s and is comparatively small after 1980. Across two poverty concepts and two data systems, the decomposition assigns the fertility dividend to the same historical window—the period when family-size decline was fastest—and the cumulative contribution is similar in both analyses.

Second, the long-run series suggests that fertility likely played an even larger role before the CPS window. The largest implied fertility contribution comes before 1950, when mean family size fell from about 4.2 to roughly 3 children—a decline at least as large as the subsequent fall to 2.4. The 1950s baby boom temporarily reversed the family-size decline: rising family size mechanically pushed child poverty upward before the secular decline resumed.

These long-run results are supported by the stability of the underlying poverty–family size gradient. From 1950 to 2023, children in families with four or more children face poverty rates roughly 2.5 times those in one-child families, and this ratio changes little despite large swings in overall poverty levels (Appendix Figure D1, Panel (b)). That the fertility component appears under both the SPM and OPM—and exhibits the same concentration in time—supports interpreting it as a robust feature of how family size maps fixed household resources into children’s material living standards, rather than an artifact of a particular poverty measure, sample period, or decomposition design.

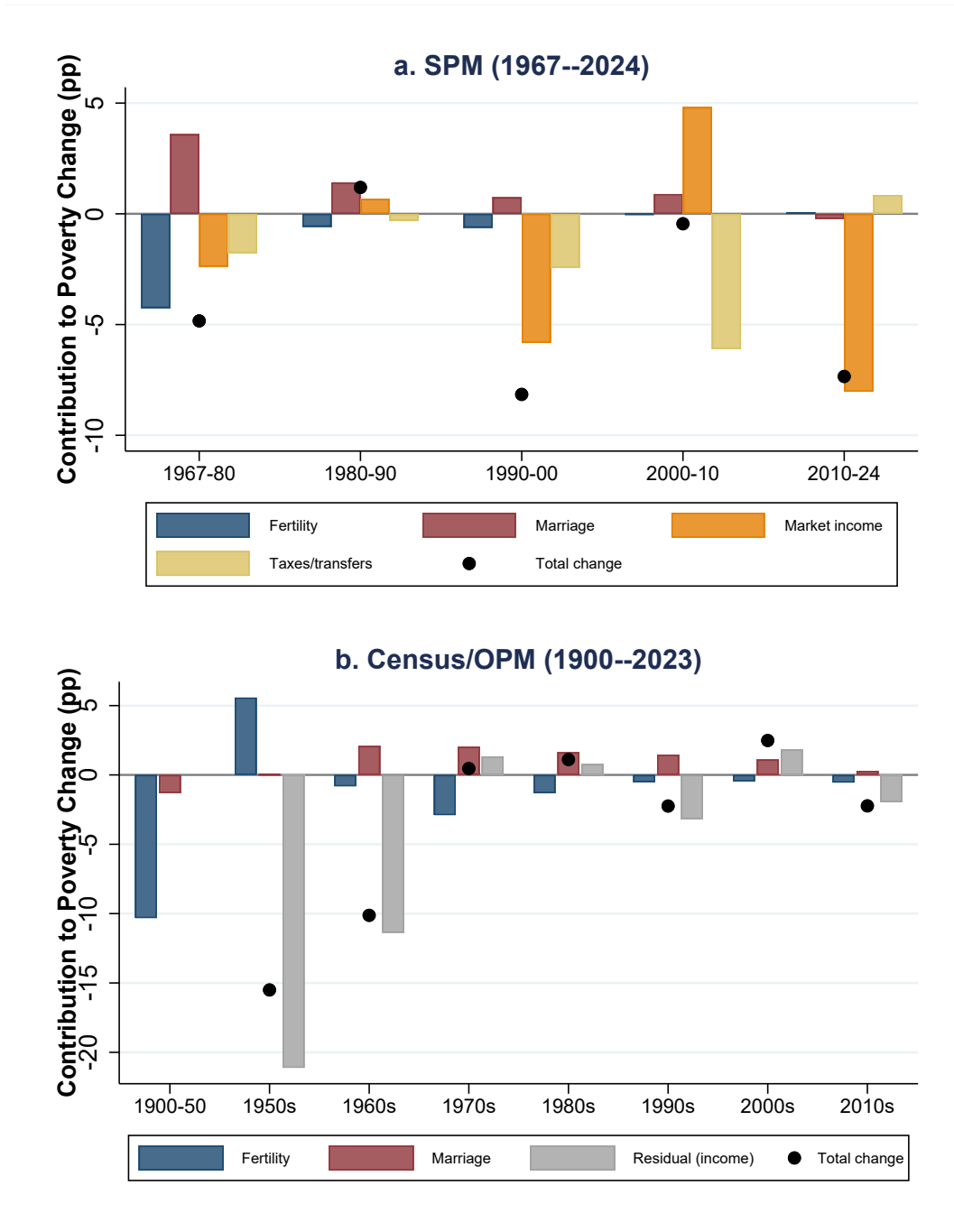


Figure 3: Fertility’s contribution to falling child poverty was concentrated before 1980

*Notes:* Panel a decomposes changes in SPM child poverty (1967–2024) into Shapley contributions from fertility, marriage, market income, and taxes/transfers. Panel b decomposes changes in OPM child poverty (1950–2023) into fertility (rank-preserving transport), marriage (IPW reweighting), and a residual capturing income and policy changes. OPM poverty is consistently available from 1950 onward; the 1900–1950 fertility and marriage bars are implied effects computed by applying backward transport to the 1950 base-line and plotting the negative—the total change and residual are not observed for this period. Black dots show total observed poverty change (omitted for 1900–1950). Sources: CPS ASEC (panel a), IPUMS USA Census/ACS (panel b).

## Conclusion

Declining family size was a substantial and overlooked contributor to the postwar fall in child poverty. Between 1967 and 2024, smaller families raised resources per child and reduced child poverty by 5.4 percentage points, accounting for 27% of the decline. Twin-birth estimates support the resource-dilution mechanism, while the 1900–2023 Census extension shows that the same channel was likely important earlier in the century.

The result sharpens child-poverty accounting in two ways. First, it shows that conventional decompositions can misclassify demographic change as income growth. Because poverty measures equalize resources and standard decompositions condition on family size, falling family size raises measured resources relative to needs without appearing as a distinct demographic contribution. Recognizing this channel reattributes part of the measured income contribution to a decline in the number of children sharing family resources. Second, the channel remains active even though its contribution to recent trends is small. The twin estimates show that an additional child raises a family’s child-poverty risk by 3–7 percentage points in current data; what ended after 1980 was the decline in family size among parents that fed it, not the mechanism itself. Today’s fertility decline operates increasingly through childlessness and fewer first births (Kearney and Levine 2025; Kearney et al. 2022), a margin that changes the number of children rather than the resources available to children with siblings. By the same token, the estimates put a number on a missing margin in current pronatalist debates: policies that raise births among existing parents could raise child poverty by roughly the magnitudes estimated here.

Finally, the findings add a child-focused perspective to debates about low fertility and population aging (e.g., Spears and Geruso 2025; Weil 2026). Lower fertility may strain old-age financing, but increasing fertility raises the number of dependents long before it raises the number of workers. That is the macroeconomic transition cost emphasized by Weil (2026); the results here show the corresponding child-level margin: during the period

when families were shrinking, fewer children raised resources per child and reduced child poverty.

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